

## Food Safety And Standards (Ayurveda Aahara) Regulations, 2022

[05 May 2022]

F. No. Stds/SP-05/A-1. Y(01).-Whereas the draft of Food Safety and Standards (Ayurveda Aahara) Regulations, 2021 were published as required by section 92 of the Food Safety and Standards Act, 2006 (34 of 2006), vide notification of the Food Safety and Standards Authority of India number F. No. Stds/SP-05/A-1. Y(01), dated the 30th June 2021, in the Gazette of India, Extraordinary, Part III, Section 4, inviting objections and suggestions from the persons likely to be affected thereby, before the expiry of the period of sixty days from the date on which the copies of the Official Gazette containing the said notification were made available to the public;

And whereas, the copies of the said Gazette were made available to the public on the 5th July, 2021;

And whereas the objections and suggestions received from the public in respect of the said draft regulations have been considered by the Food Safety and Standards Authority of India;

Now, therefore, in exercise of the powers conferred by clause (v) of sub-section (2) of section 92 of the said Act, the Food Safety and Standards Authority of India with the previous approval of Central Government, hereby makes the following regulations, namely.

### Regulation - 1.

- (1) These regulations may be called as the Food Safety and Standards (Ayurveda Aahara) Regulations, 2022.
- (2) They shall come into force on the date of their publication in the Official Gazette.

### Regulation - 2. Definitions.

- (a) "Act" means the Food Safety and Standards Act, 2006 (34 of 2006);
- (b) "Ayurveda Aahara" means a food prepared in accordance with the recipes or ingredients or processes as per method described in the authoritative books of Ayurveda listed under Schedule A of these regulations including products which have other botanical ingredients in accordance with the concept of Ayurveda Aahara but does not include Ayurvedic drugs or proprietary Ayurvedic medicines and medicinal products, cosmetics, narcotic or psychotropic substances, herbs listed under Schedule E-1 of Drug and Cosmetics Act, 1940 and the Drug and Cosmetics Rules, 1945, metals based Ayurvedic drugs or medicines, bhasma or pishti and any other ingredients notified by the Authority from time to time.

Explanation 1.- Food is defined under the Food Safety and Standards Act, 2006 (34 of 2006) and regulations made thereunder. Recipes and ingredients specified in the Schedule A authoritative books for promoting health or to meet specific physiological needs, including those foods specified for consumption during or post specified diseases, disorders referred as pathya in Ayurveda are covered under these regulations.

Explanation 2.- Adoption of processes for cooking or preparation of Ayurveda Aahara specified or described in the Schedule A authoritative books for industrial scale manufacture and packing are permitted. Such adoptions shall be aimed to produce Ayurveda Aahara with quality and characteristics closely similar as specified in the books mentioned under Schedule A of these regulations.

Explanation 3.- Any packed food item used in day to day life for dietary purposes such as pulses, rice, flour, vegetable including other foods specified under Food Safety and Standards Regulations shall not be covered under these regulations unless otherwise they meet the provisions of these regulations.

Minimally processed food items involving cleaning, polishing, dehusking, grading shall not fall under these regulations.

(c) "Food Authority" means the Food Safety and Standards Authority of India established under section 4 of the Act;

(d) "Schedule" mean a schedule to these regulations.

### **Regulation - 3. General requirements.**

- (1) Food Business Operators shall formulate Ayurveda Aahara in accordance with the categories and requirements specified in Schedule B of these regulations.
- (2) No person shall manufacture or sell Ayurveda Aahara intended for administration to infants up to the age of twenty-four months.
- (3) The Food Business Operator shall manufacture Ayurveda Aahara in accordance with the Schedule 4 of the Food Safety and Standards (Licensing and Registration of Food Businesses) Regulation, 2011.
- (4) No person shall add vitamins, minerals and amino acids to Ayurveda Aahara:

Provided that natural vitamins and minerals if present in the Ayurveda Aahara may be declared on the label.

- (5) Food Business Operator may adopt the purity criteria for the ingredients as per the Food Safety and Standards Regulations or generally accepted by pharmacopoeias (namely, Indian Pharmacopoeia, Ayurvedic Pharmacopoeia of India, British Pharmacopoeia, United States Pharmacopoeia), or relevant BIS Specifications, or Quality Standards of Indian Medicinal Plants as published by Indian Council of Medical Research. The Food Business Operator shall provide information on the purity criteria adopted for ingredients at the time of licensing and any subsequent changes.

### **Regulation - 4. Additives.**

The products covered under these regulations shall contain only food additives as specified in Schedule C of these regulations.

### **Regulation - 5. Contaminants.**

The products covered under these regulations shall conform to the safety requirements specified in Schedule D of these regulations.

### **Regulation - 6. Packaging.**

Ayurveda Aahara shall conform to the Food Safety and Standards (Packaging) Regulations, 2018.

### **Regulation - 7. Restriction on sale, manufacture, etc. of Aurveda Ahara.**

No person shall manufacture, pack, sell, offer for sale, market or otherwise distribute or import Ayurveda Aahara unless the product complies with the requirements laid down in these regulations.

### **Regulation - 8. Not to claim treating human disease, etc.**

The labelling, presentation and advertisement shall not claim that the Ayurveda Aahara has the property of preventing, treating or curing a human disease or refer to such properties.

### **Regulation - 9. Claim by Food Business Operator.**

- (1) Food Business Operator shall make claims in accordance with the Food Safety and Standards (Advertising and Claims) Regulations, 2018.
- (2) Health claims and Disease risk reduction claims for the different categories of Ayurveda Aahara and their approval process shall be in accordance with the requirements specified in Schedule B of these regulations.

### **Regulation - 10. Prior approval for Ayurveda Aahara.**

If Ayurveda Aahara need prior approval as per Schedule B of these regulations, the same shall be in accordance with the Food Safety and Standards (Approval for Non-Specific Food and Food Ingredients) Regulation, 2017.

### **Regulation - 11. Constitution of Expert committee.**

The Food Safety and Standards Authority of India shall constitute an Expert Committee under the Ministry of Ayush consisting of relevant experts including representatives of Food Safety and Standards Authority of India for providing recommendation on approval of claims and products as specified in sub-regulation (9) and (10) and the Committee shall also empower to address concerns regarding registration or licensing or certification or laboratory accreditation or testing or quality issues related to Ayurveda Aahara.

### **Regulation - 12. Labelling of Ayurveda Aahara.**

The labelling of Ayurveda Aahara shall be in accordance with the Food Safety and Standards (Labelling and Display) Regulations, 2020, and the specific labelling requirements provided in these regulations.

### **Regulation - 13. Other requirement for Ayurveda Aahara.**

Every label of Ayurveda Aahara shall specify the intended purpose, the target consumer group, recommended duration of use and other specific labelling requirements, namely.

- (a) the words "AYURVEDA AAHARA" printed in the immediate proximity of the name or brand name of the product; and the logo as specified in the Schedule E of these regulations, on the front of the pack of the label;
- (b) an advisory warning ONLY FOR DIETARY USE prominently written;
- (c) a statement that the Ayurveda Aahara is not to be used as a substitute for a varied diet;
- (d) a warning or any other precautions to be taken while consuming, known side effects, if any, contraindications, and published product or drug interactions, as applicable; and.
- (e) a statement that the product is required to be stored out of reach of children.
- (f) a warning that the product is for oral consumption only and not for parenteral use.

### **Regulation - 14. Information by the Food Business Operator to Food Authority.**

The Food Business Operator shall inform the licensing authority in writing, if any, of his existing food products duly licensed to be assigned as an Ayurveda Aahara and the Licensing Authority shall within thirty days from the date of recommendation received from the Expert Committee, permit the same with applicable modifications including labelling as specified in these regulations.

Schedule A  
[See regulation 2(b)]

List of authoritative books\* for Ayurveda Aahara.

| Sl. No. | Books                                                 |
|---------|-------------------------------------------------------|
| 1.      | Abhinava Chintamani                                   |
| 2.      | Arka Prakasha                                         |
| 3.      | Arogya Kalpadruma                                     |
| 4.      | Arya Bishak                                           |
| 5.      | Ashtanga Hridaya                                      |
| 6.      | Ashtanga Samgraha                                     |
| 7.      | Ayurveda Chintamani                                   |
| 8.      | Ayurveda Kalpadruma                                   |
| 9.      | Ayurveda Prakasha                                     |
| 10.     | Ayurveda Ratnakara                                    |
| 11.     | Ayurveda Samgraha                                     |
| 12.     | Bangasena                                             |
| 13.     | Ayurvedic Formulary of India                          |
| 14.     | Ayurveda Sara Samgraha                                |
| 15.     | Ayurvedic Pharmacopeia of India                       |
| 16.     | Bhaishajya Ratnavali                                  |
| 17.     | Bhava Prakash                                         |
| 18.     | Bhela Samhita                                         |
| 19.     | Bhojana Kutuhalam                                     |
| 20.     | Brihat Bhaishajya Ratnakar                            |
| 21.     | Brihat Nighantu Ratnakar                              |
| 22.     | Chakra Datta                                          |
| 23.     | Charak Samhita                                        |
| 24.     | Dravyaguna Nighantu                                   |
| 25.     | Gada Nigraha                                          |
| 26.     | Harama khala                                          |
| 27.     | Kaideva Nighantu                                      |
| 28.     | Kashyapa Samhita                                      |
| 29.     | Ksemakutuhalam                                        |
| 30.     | Kupi Pakva Rasayana                                   |
| 31.     | Madanpala Nighantu                                    |
| 32.     | Manasollasa / Abhilashitarah Chintamani               |
| 33.     | Nighantu Ratnakar                                     |
| 34.     | Paka Darpana of Nala                                  |
| 35.     | Pathya Apathya Vinishchaya                            |
| 36.     | Raja Nighantu                                         |
| 37.     | Rasa Chandanshu                                       |
| 38.     | Rasa Pradipika                                        |
| 39.     | Rasa Raja Sundara                                     |
| 40.     | Rasa Ratna Samuchchaya                                |
| 41.     | Rasa Tantra Sara va Siddha Prayaoga Sangraha - Part 1 |
| 42.     | Rasa Tarangini                                        |
| 43.     | Rasa Yoga Ratnakar                                    |
| 44.     | Rasa Yoga Sagara                                      |
| 45.     | Rasa Yoga Samgraha                                    |
| 46.     | Rasamanjiri                                           |
| 47.     | Rasamrita                                             |
| 48.     | Rasendra Sara Samgraha                                |
| 49.     | Ruchivadhu Gala Ratnamala                             |
| 50.     | Sahasrayoga                                           |
| 51.     | Sarvaroga Chikitsa Ratnam                             |
| 52.     | Sarvayoga Chikitsa Ratnam                             |
| 53.     | Sharangdhara Samhita                                  |
| 54.     | Shodhala Nighantu                                     |
| 55.     | Siddha Bhaishajya Manimala                            |
| 56.     | Siddha Yoga Samgraha                                  |
| 57.     | Siva Tatva Ratnakara II                               |
| 58.     | Soopa Shastra of Mangarasa III                        |
| 59.     | Susena Nighantu/Ayurveda Mahodadhi                    |
| 60.     | Sushruta Samhita                                      |
| 61.     | Vaidya Chintamani                                     |
| 62.     | Vaidyaka Chikitsa Sara                                |
| 63.     | Vaidyaka Shabda Sindhu                                |
| 64.     | Vasava Rajeeyam                                       |
| 65.     | Vidya Jiwan                                           |
| 66.     | Vishwanath Chikitsa                                   |
| 67.     | Vrinda Chikitsa                                       |
| 68.     | Yoga Chintamani                                       |
| 69.     | Yoga Ratnakara                                        |
| 70.     | Yoga Tarangini                                        |
| 71.     | Yogaratna Samgraha                                    |

Note 1.- Ingredients and recipes listed in the above editions of authoritative books and those authoritative texts published before 1940 shall be considered by the Food Authority for approval as

## Ayurveda Aahara.

Note 2.- Ingredients and recipes listed in Appendix or Annexure to the above authoritative texts in Schedule A shall not be considered by the Food Authority for approvals Ayurveda Aahara.

## Schedule B

[See regulation 3 (1)]

## Categories of Ayurveda Aahara and Regulatory Requirements.

| Category | Description                                                                                                                                                                                                                                                     | Ingredients Permitted and Process                                                                                               | Safety Data (prior approval of product)                                                                                                                                                           | Label Claim                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| (1)      | (2)                                                                                                                                                                                                                                                             | (3)                                                                                                                             | (4)                                                                                                                                                                                               | Health Benefit (5) Disease Risk Reduction (6)                                                                                                   |
| A        | Ayurveda Aahara prepared in accordance with the Authoritative Ayurveda Texts in Books listed in Schedule A.                                                                                                                                                     | As provided in the Authoritative Ayurveda Texts in Books listed in Schedule A                                                   | Not Required                                                                                                                                                                                      | Prior approval of the Food Authority not required if claim is as per the Authoritative Ayurveda Texts in Books listed in Schedule A             |
| B        | A new recipe of Ayurveda Aahara using ingredients listed in the Authoritative Ayurveda Texts in Books listed in Schedule A, along with other botanicals used in Ayurveda Dietetic Principles (viz. Rasa , Guna, Virya Vipaka, and Karma)                        | Text Reference required for using new recipe or ingredient(s) provided in the Authoritative Ayurveda Books listed in Schedule A | Required (Rationale, and Safety Data (including mutagenicity, carcinogenicity, and teratogenicity testing) of the new recipe or ingredient(s) or supporting published modern scientific evidence) | Prior approval required based on the Authoritative Ayurveda Texts in Books listed in Schedule A                                                 |
| B1       | Ayurveda Aahara presented in a format different from that specified in the Authoritative Texts of Books listed in Schedule A                                                                                                                                    | As provided in the Authoritative Ayurveda Texts in Books listed in Schedule A (See explanatory note below)                      | Not Required                                                                                                                                                                                      | Rationale and Efficacy Data of new format including the target population, required for prior approval of the Food Authority                    |
| B2       | Ayurveda Aahara intended to provide a health benefit, or as an adjuvant, to support a specific disease condition, or disorder specified/ not specified in the Authoritative Texts of Books listed in Schedule A. (Ayurveda Aahara for Specific Medical Purpose) | As provided in the Authoritative Ayurveda Texts in Books listed in Schedule A                                                   | Not Required                                                                                                                                                                                      | Rationale and Efficacy Data for the specific medical purpose including the target population, required for prior approval of the Food Authority |

Note 1: The Food Authority may provide a list of Ayurveda Aahara covered under category A above from time to time. Addition or deletion or modification to this list shall be informed to the public by the Authority from time to time. Food Business Operator manufacturing such products may approach licensing Authority without need of any prior approval of the Authority. In case of Ayurveda Aahara products falling under category A, but not mentioned in the list, Food Business Operator shall request the Food Authority for inclusion by submitting relevant literature from Authoritative text listed under Schedule A books.

Note 2: The mere adoption of the Ayurveda Aahara in a format for suitability to offer it in a pre-packaged condition including concentration and drying for reconstitution prior to use may not constitute a change in format.

## Schedule C

[See regulation 4]

## List of Additives permitted in Ayurveda Aahara

| S.No. | Food Additive                                                                                  | Maximum permitted Level | Functional classes    |
|-------|------------------------------------------------------------------------------------------------|-------------------------|-----------------------|
| 1.    | Guar Arabic/ Acacia gum (INS414)                                                               | 2%                      | Thickener, stabilizer |
| 2.    | Tragacanth gum                                                                                 | 2%                      |                       |
| 3.    | Guar gum (INS 412)                                                                             | 2%                      |                       |
| 4.    | Pectins (INS 440)                                                                              | GMP*                    | Sweetening agent      |
| 5.    | Gum Karaya (INS 416)                                                                           | GMP*                    |                       |
| 6.    | Konjac flour (INS 425)                                                                         | GMP*                    |                       |
| 7.    | Starch and starch derivatives permitted in FSSR                                                | GMP*                    |                       |
| 8.    | Honey                                                                                          | GMP*                    |                       |
| 9.    | Jaggery                                                                                        | GMP*                    | Colour                |
| 10.   | Date syrup                                                                                     | GMP*                    |                       |
| 11.   | Mollasses                                                                                      | 5%                      |                       |
| 12.   | Curcumin (INS 100 (i))                                                                         | GMP*                    | Colour                |
| 13.   | Turmeric (INS 100 (ii))                                                                        | GMP*                    |                       |
| 14.   | Paprika/ Paprika Extract/ Paprika Oleoresin (INS 160C (ii))                                    | GMP*                    |                       |
| 15.   | Annatto extract, norbixin-based(INS 160b(ii))                                                  | 100 mg/kg               | Acidity regulator     |
| 16.   | Annatto extract, bixin-based (INS 160b(ii))                                                    | 50 mg/kg                |                       |
| 17.   | Chlorophyll A and Chlorophyll B(INS 140)                                                       | 100 mg/kg               |                       |
| 18.   | Anthocyanins (INS 163)                                                                         | GMP*                    | Acidity regulator     |
| 19.   | Caramel plain (150a)                                                                           | GMP*                    |                       |
| 20.   | Concentrated water extract of a colored fruit or vegetable listed under FSSR (Colouring foods) | GMP*                    |                       |
| 21.   | Acetic acid (INS 260)                                                                          | GMP*                    | Acidity regulator     |
| 22.   | DL-Lactic acid (INS 270)                                                                       | GMP*                    |                       |
| 23.   | Citric acid (INS 330)                                                                          | GMP*                    |                       |
| 24.   | Tartaric acid (INS 334)                                                                        | GMP*                    | Acidity regulator     |
| 25.   | DL-Malic acid (INS 296)                                                                        | GMP*                    |                       |
| 26.   | L-Ascorbic acid (INS 300)                                                                      | GMP*                    |                       |
| 27.   | Rose oil                                                                                       | 1%                      | Flavouring agent      |
| 28.   | Kewda/Kewra (oil from Ketaki)                                                                  | 0.5%                    |                       |
| 29.   | Rosemary oil                                                                                   | 1%                      |                       |
| 30.   | Distilled oils of spices                                                                       | 2%                      | Antioxidant           |
| 31.   | Powders of spices                                                                              | 1%                      |                       |
| 32.   | Nibu satva (citric acid)                                                                       | GMP*                    |                       |

Quantity of the additive added to food shall be limited to the lowest possible level necessary to accomplish its desired effect.

#### Schedule D [See regulation 5]

#### Microbiological Standards for Ayurveda Aahara

Table 1A. Process Hygiene Criteria.

| Sr.No. | Product description                                       | Aerobic Plate Count (cfu/g or ml) |   |       |     | Yeast and Mould Count (cfu/g or ml)   |   |       |     | Enterobacteriaceae count(cfug or ml) |   |       |     |
|--------|-----------------------------------------------------------|-----------------------------------|---|-------|-----|---------------------------------------|---|-------|-----|--------------------------------------|---|-------|-----|
|        |                                                           | Sampling plan                     |   | Limit |     | Sampling plan                         |   | Limit |     | Sampling plan                        |   | Limit |     |
| 1.     | Ayurveda Aahara unprocessed and not for directconsumption | n                                 | c | m     | M   | n                                     | c | m     | M   | n                                    | c | m     | M   |
| 2.     | Ayurveda Aahara for direct consumption                    | 5                                 | 3 | 106   | 107 | 5                                     | 3 | 104   | 105 | 5                                    | 3 | 103   | 104 |
| 3.     | Ayurveda Aahara fermented products*                       | 5                                 | 2 | 104   | 105 | 5                                     | 2 | 102   | 103 | 5                                    | 2 | 102   | 103 |
| 4.     | Test Methods                                              | IS: 5402/ISO:4833                 |   |       |     | IS: 5403/ ISO 21527 Part 1 and Part 2 |   |       |     | IS/ISO 7402/ ISO 21528 Part 2        |   |       |     |

Table 1B. Food Safety Criteria

| Sr.No. | Product description                                      | Salmonella                |    |            |            | Listeria monocytogenes          |    |            |            |
|--------|----------------------------------------------------------|---------------------------|----|------------|------------|---------------------------------|----|------------|------------|
|        |                                                          | Sampling plan             |    | Limit      |            | Sampling plan                   |    | Limit      |            |
| 1.     | Ayurveda Aahara unprocessed and notfor directconsumption | n                         | c  | m          | M          | n                               | c  | m          | M          |
| 2.     | Ayurveda Aahara for direct consumption                   | NA                        | NA | NA         | NA         | NA                              | NA | NA         | NA         |
| 3.     | Ayurveda Aahara fermented products*                      | 5                         | 0  | Absent/25g | Absent/25g | 5                               | 0  | Absent/25g | Absent/25g |
| 4.     | Test Methods                                             | IS: 5887 Part3 / ISO:6579 |    |            |            | IS: 14988, Part 1 / ISO 11290-1 |    |            |            |

Should contain only the specified microorganism at the level claimed on the label. The counts have to be determined using methodologyappropriate for the organisms, for example for Lactic acid bacteria ISO 15214/IS 16068, for Bifidobacteria ISO29981

Note.- In high value low volume (less than 100 g) and large retail pack (pack more than 1 kg) sizes, the sample plan may be modified (eg.absence of Salmonella in 10g or 5g in the case of former or n number of samples to be taken from different sites of one large pack) accordingly on case to case basis with the prior approval of Food Safety and Standards Authority of India.

Stage where the Microbiological Standards shall apply.- The microbiological standards with respect to the products categories specified as Process Hygiene Criteria indicate the acceptable functioning of the production process. These are not to be used as requirements for releasing the products in the market.

These are indicative values above which corrective actions are required in order to maintain the hygiene of the process in compliance with food law. These shall be applicable at the end of the manufacturing process. The Microbiological Standards as Food Safety Criteria define the acceptability of a batch/lot and shall be met in respect of the products at the end of the manufacturing process and the products in the market during their shelf- life.

Action in case of unsatisfactory result.- In case of non-compliance in respect of process hygiene criteria, the Food Business Operator shall.

check and improve process hygiene by implementation of guidelines in Schedule 4 of FSS (Licensing and Registration of Food Businesses) Regulations; and,

ensure that all food safety criteria are complied with. Sampling Plan and Guidelines

Guidelines for Regulator: The sampling for different microbiological standards specified in Table 1B shall be ensured aseptically at manufacturing units and at retail points, as applicable, by a trained person with specialized knowledge in the field of microbiology following guidelines in the Food Safety and Standards (Food Products and Food Additives) Regulations, 2011 and ISO: 17728. The samples shall be stored and transported in frozen condition at  $-18^{\circ}\text{C}(\pm 2^{\circ}\text{C})$  or under refrigerated conditions at  $2-5^{\circ}\text{C}$  as applicable except for the products that are recommended to be stored at room temperature by the manufacturer to enable initiation of analysis within twenty four hours of sampling.

Preservatives shall not be added to sample units intended for microbiological examination. The desired number of sample units as per sampling plan given in Table 1B shall be taken from same batch or lot and shall be submitted to the notified laboratories. Three sets, each containing n number of samples (n as defined in the sampling plan eg if  $n=5$ , then total no. of samples to be drawn is 15) shall be drawn.

Each of these three sets shall be tested in three different accredited laboratories. The final decision shall be based on the results of three accredited laboratories. In the case of Food Safety Criteria, the results from all the three laboratories should indicate compliance with the specified criteria. There shall be no provision for retesting or resampling for microbiological testing. The testing in laboratory shall be ensured as per the methods given in the table "reference test methods"

Guidelines for Food Business Operator: The Food Business Operator shall perform testing as appropriate as per the microbiological standards specified in the Tables to these regulations above to ensure verification of compliance with the microbiological requirements. The Food Business Operator shall decide themselves subject to minimum prescribed under Food Safety and Standards (Licensing and Registration of Food Businesses) Regulation, 2011, the necessary sampling and testing frequencies to ensure compliance with the specified microbiological requirements. The Food Business Operator may use analytical methods other than those described in "reference test methods" given below for inhouse testing only. However, these methods shall not be applicable for regulatory compliance purpose.

Sampling Plan:

The terms n,c,m and M used in this standard have the following meaning:

n = Number of units comprising a sample.

c = Maximum allowable number of units having microbiological counts above m for 2- class sampling

plan and between m and M for 3- class sampling plan.

m = Microbiological limit that separates unsatisfactory from satisfactory in a 2- class sampling plan or acceptable from satisfactory in a 3-class sampling plan.

M = Microbiological limit that separates unsatisfactory from satisfactory in a 3-class sampling plan.

#### Interpretation of Results:

| 2-Class Sampling Plan (where n,c and m are specified)          | 3-Class Sampling Plan (where n,c,m and M are specified)                                               |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1. Satisfactory, if all the values observed are m              | 1. Satisfactory, if all the values observed are m                                                     |
| 2. Unsatisfactory, if one or more of the values observed are m | 2. Acceptable, if a maximum of c values are between m and M.                                          |
|                                                                | 3. Unsatisfactory, if one or more of the values observed are M or more than prescribed c values are m |

Reference test methods.- The following test methods shall be applied as reference methods. Test methods prescribed in Food Safety and Standards Authority of India Manual of Method of Analysis of Foods (Microbiological Testing) may also be referred along with the IS/ISO methods specified for Process Hygiene Criteria and Food Safety Criteria. Latest version of test methods shall apply. In case where an ISO method adopted by the BIS is specified (e.g. IS XXXX / ISO YYYY), latest version of the ISO method (or its BIS equivalent, if available) shall apply.

| S.No | Parameter                | Reference Test methods                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Aerobic Plate Count      | Microbiology of the food chain -- Horizontal method for the enumeration of microorganisms -- Part 1: Colony count at 30 °C by the pour plate technique- IS 5402/ ISO:4833                                                                                                                                                                                                                                                                                                      |
| 2.   | Yeast and Mold Count     | Method for Yeast and Mold Count of Food Stuffs and Animal feed- IS 5403 Microbiology of food and animal feeding Stuff-Horizontal method for the enumeration of yeasts and moulds-Part1: Colony count technique in products with water activity greater than 0.95-ISO 21527-1 Microbiology of food and animal feeding Stuff-Horizontal method for the enumeration of yeasts and moulds-Part2: Colony count technique in products with water activity less than 0.95-ISO 21527-2 |
| 3.   | Enterobacteriaceae count | Microbiology - General Guidance for the Enumeration of Enterobacteriaceae without Resuscitation - MPN Technique and Colony-count Technique- IS/ISO 7402 Microbiology of Food and Animal feeding stuff -Horizontal methods for the detection and enumeration of Enterobacteriaceae- Part 2: Colony- count method-ISO 21528-2                                                                                                                                                    |
| 4.   | Salmonella               | Methods for Detection of Bacteria Responsible for Food Poisoning - Part 3: General Guidance on Methods for the Detection of Salmonella- IS 5887: Part 3 Microbiology of food and animal feeding stuffs -- Horizontal method for the detection of Salmonella spp.- ISO 6579                                                                                                                                                                                                     |
| 5.   | Listeria monocytogenes   | Microbiology of the food chain -- Horizontal method for the detection and enumeration of Listeria monocytogenes and of Listeria spp. -- Part 1: Detection method -ISO 11290-1 Microbiology of Food and Feeding Stuffs - Horizontal method for Detection and Enumeration of Listeria Monocytogenes, Part 1: Detection Method -IS 14988-1                                                                                                                                        |

Table 2: Permissible levels of Contaminants.

| Name of contaminant                                               | Limits (Maximum) |
|-------------------------------------------------------------------|------------------|
| Lead, mg per kg or mg per L                                       | 2.5              |
| Copper, mg per kg or mg per L                                     | 30               |
| Arsenic, mg per kg or mg per L                                    | 1.1              |
| Tin, mg per kg or mg per L                                        | 250              |
| Cadmium, mg per kg or mg per L                                    | 1.5              |
| Mercury, mg per kg or mg per L                                    | 1.0              |
| Methyl Mercury (Calculated as the element), mg per kg or mg per L | 0.25             |
| Total Aflatoxins*, µg per kg                                      | 20               |
| Aflatoxin B1*, µg per kg                                          | 10               |
| Melamine, mg per kg                                               | 2.5              |

(Food products containing Arecanut or Betelnut, cereal and cereal products, dried figs, nuts (for further processing and ready to eat), oilseeds or oil (for further processing and ready to eat), pulses and spices/spice mix)